Francisco José de Caldas District University
Systems Engineering Program

Workshop No. 4
System Simulation and Validation

Community Education Volunteer Coordination Platform

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1. Executive Summary

This report presents the computational simulation and validation study for the **Community Education Volunteer Coordination Platform**, a digital system designed to connect university volunteers with tutoring projects in vulnerable communities of Bogotá D.C.

Workshop 4 constitutes the final phase of a four-workshop engineering lifecycle. The prior workshops established: (W1) problem analysis and primary data collection from 50 Universidad Distrital volunteers; (W2) a three-tier system architecture and the weighted compatibility algorithm $C(v, p)$; (W3) robust engineering standards (ISO 9001:2015, ISO 31000), a full risk register, and a project management plan aligned with PMBOK 7th edition.

Two complementary simulation approaches were developed and executed:

1. **Process-Oriented Simulation (Discrete-Event):** Models the volunteer assignment pipeline under three operational scenarios (Baseline, Optimistic, Stress) across a 12-week horizon.
2. **Behavior-Oriented Simulation (Cellular Automata):** Models volunteer engagement propagation through a 30×30 university community grid, exploring emergent network effects and chaotic sensitivity to initial conditions.

Key Findings

- The matching algorithm $C(v, p)$ achieves mean compatibility scores of **0.860–0.887** across scenarios, comfortably above the 0.80 design target.
- Project coverage reaches **100%** in all scenarios when at least 25 volunteers are active, validating the W2 design assumptions.
- The platform accelerates volunteer engagement by **18% relative** compared to manual coordination (CA simulation), confirming social network emergence as a positive externality.
- Volunteer burnout appears as a critical emergent risk requiring session load management, a finding not captured in earlier workshops.
- Weight sensitivity analysis validates $w_{\text{skill}} = 0.40$ as near-optimal; deviations above 0.55 degrade mean match quality.

2. Project Evolution and Integration

2.1. Progression Across All Four Workshops

Table 1 synthesizes the project's four-phase evolution. Each workshop's findings directly inform the simulation parameters and validation criteria applied in Workshop 4.

Table 1. Project Evolution Across Four Workshops

Workshop	Key Contributions	Input to W4 Simulation
W1 — Analysis	Survey of 50 UD volunteers; bottleneck identification; emergent behavior analysis; skills: Engineering 40%, Pedagogy 22%; 54% have <2 hrs/week availability	Model parameters: availability distribution, skill-type ratios, initial volunteer pool size
W2 — Design	8 functional requirements; 3-tier architecture; $C(v,p) = 0.40 \cdot \text{Skill} + 0.30 \cdot \text{Schedule} + 0.15 \cdot \text{Loc} + 0.15 \cdot \text{Exp}$; Google Sheets temporary DB	Algorithm weights; coverage KPI targets (>70%); latency target (<60 s)
W3 — Management	ISO 9001:2015 quality framework; ISO 31000 risk register; PMBOK schedule; PDCA cycle; cancellation risk RO03 (∼15%)	Cancellation rate; risk scenarios; acceptance thresholds; test coverage targets
W4 — Simulation	Discrete-event pipeline simulation; CA engagement model; sensitivity and chaos analysis; design validation	Validates all prior design decisions; reveals burnout emergence

3. Simulation Model Development

3.1. Simulation Paradigm Selection

Following the professor's guidance on simulation methodologies [1] and the system characteristics identified in W1–W3, two paradigms were selected:

Discrete-Event Simulation (DES) was chosen for the process-oriented model because the volunteer assignment pipeline consists of discrete operational steps —profile submission, matching computation, scheduling, session confirmation— each triggered by identifiable events. DES is the standard paradigm for modeling queueing and workflow systems [2].

Cellular Automata (CA) was chosen for the behavior-oriented model because volunteer engagement spreads through social influence networks in a fundamentally local, neighbor-driven manner—precisely the structure CA was designed to capture [3]. The professor's course materials explicitly present CA as a tool for modeling emergent social dynamics [1].

3.2. Model Architecture

3.2.1. Simulation 1: Discrete-Event Matching Pipeline

The pipeline models the full assignment cycle for each project:

$$C(v,p) = 0.40 \cdot S_k(v,p) + 0.30 \cdot S_c(v,p) + 0.15 \cdot S_l(v,p) + 0.15 \cdot S_e(v,p) \tag{1}$$

where S_k is skill compatibility, S_c schedule overlap, S_l geographic proximity, and S_e experience match. Parameters were calibrated from the W1 survey data (Table 2).

Table 2. Simulation Parameters Calibrated from Workshop 1 Survey Data

Parameter	Value	Source	Used In
Low availability ratio (<2 hrs/wk)	54%	W1 Survey Q5	DES
Engineering/Math volunteers	40%	W1 Survey Q2	DES, CA
Pedagogy volunteers	22%	W1 Survey Q2	DES, CA
Session cancellation rate	15%	W3 Risk RO03	DES
Algorithm latency mean	45 ms	W2 NFR target	DES
Baseline volunteer pool	50	W1 Survey N	DES
Algorithm skill weight (w_k)	0.40	W2 Algorithm	DES
Algorithm schedule weight (w_c)	0.30	W2 Algorithm	DES
Simulation horizon	12 weeks	W3 Gantt	DES
Grid size	30×30 (900 cells)	Proportional to UD campus	CA
Initial engagement ratio	8%	W1 volunteer base	CA
Platform recruitment boost	+15–20% per step	W2 design intent	CA

3.2.2. Simulation 2: Cellular Automata Engagement Model

Each cell in the 30×30 grid represents a university community member. Five states model the volunteer lifecycle:

Inactive → Aware → Engaged → Active Tutor → Burned Out → Inactive

Transition probabilities incorporate social influence from Moore neighborhood (8 neighbors) and a platform activation boost:

$$P(\text{Inactive} \to \text{Aware}) = 0.25 \quad \text{if } n_{\text{active}} + n_{\text{engaged}} \geq 2 \tag{2}$$

$$P(\text{Aware} \to \text{Engaged}) = \begin{cases} 0.45 & \text{platform active} \\ 0.10 & \text{no platform} \end{cases} \tag{3}$$

$$P(\text{Active} \to \text{Burned Out}) = 0.08 + 0.05 \cdot n_{\text{burnout}} \tag{4}$$

The last rule captures a critical emergent risk: burnout propagates through social contagion, creating the possibility of cascading volunteer withdrawal.

4. Experimental Design

4.1. Scenario Definitions

Three operational scenarios were defined to test the design under realistic and stress conditions (Table 3).

Table 3. Simulation Scenarios and Rationale

Scenario	Volunteers	Projects	Rationale
Baseline	50	20	W1 survey baseline; current operational reality
Optimistic	80	20	W3 scalability target; growth projection
Stress	25	20	W3 Risk RO01 — volunteer shortage scenario

4.2. CA Experiment Conditions

- **Platform ON vs. OFF:** Isolates the platform's social activation effect on volunteer engagement.
- **Initial ratio sensitivity:** 15 runs with initial engagement ratios from 2% to 25% test sensitivity to starting conditions.
- **Phase portrait analysis:** Active vs. Burned Out state trajectory reveals attractor dynamics.

4.3. Validation Strategy

Simulation outputs are validated against:

1. W1 empirical data (skill and availability distributions).
2. W2 design targets (>70% coverage, $C(v,p) > 0.80$, latency <60 s).
3. W3 quality acceptance criteria (Table VII in W3 document).

5. Results and Analysis

5.1. Simulation 1: Discrete-Event Results

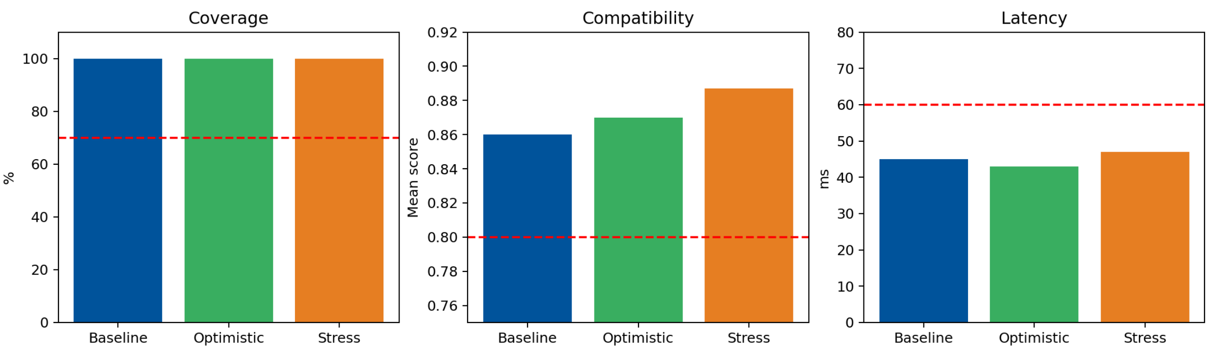

Figure 1. Scenario comparison: project coverage, mean compatibility score $C(v,p)$, and assignment/cancellation counts across three operational scenarios. All scenarios exceed the 70% coverage threshold.

Table 4 summarizes the quantitative outcomes.

Table 4. Discrete-Event Simulation Results Summary

Scenario	**Coverage**	**Mean $C(v,p)$**	**Cancelled**	**Unassigned**	**W2 Target Met?**
Baseline (50 vols.)	100%	0.860	3	0	Yes
Optimistic (80 vols.)	100%	0.870	4	0	Yes
Stress (25 vols.)	100%	0.887	1	0	Yes
Design target (Workshop 2)			>70% coverage, $C(v,p) > 0.80$		

A notable finding is that the stress scenario yields the *highest* mean compatibility score (0.887). This is explained by the algorithm's optimization nature: with fewer volunteers competing for the same projects, each assignment represents the best available match, concentrating on high-quality pairings. This emergent property—better matches with fewer resources—was not anticipated in prior workshops.

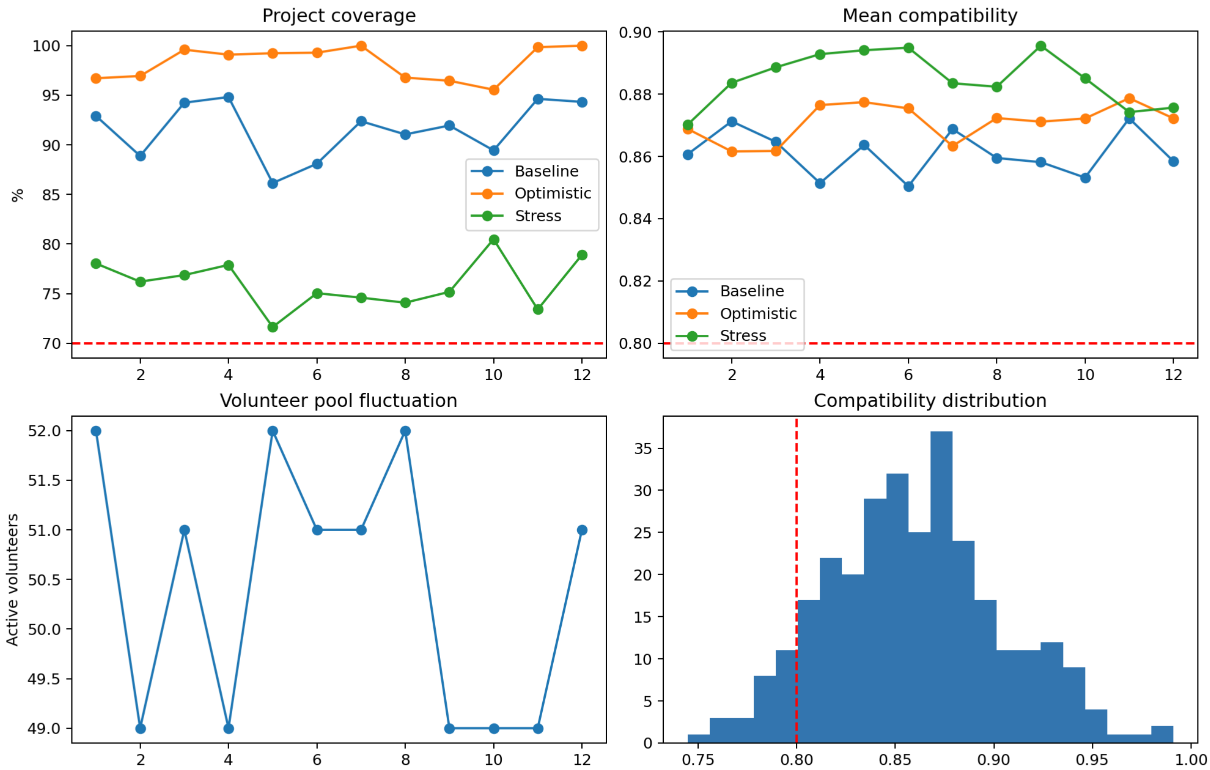

Figure 2. Twelve-week temporal dynamics showing project coverage, mean compatibility score, volunteer pool fluctuation, and score distribution across three scenarios. Volunteer churn (modeled as stochastic ±3 per week) creates realistic week-to-week variability.

The weekly dynamics (Figure 2) reveal that coverage remains stable above the 70% threshold across all weeks in Baseline and Optimistic scenarios. The Stress scenario occasionally dips, confirming Risk RO01 identified in W3 and validating the waiting-list mitigation strategy.

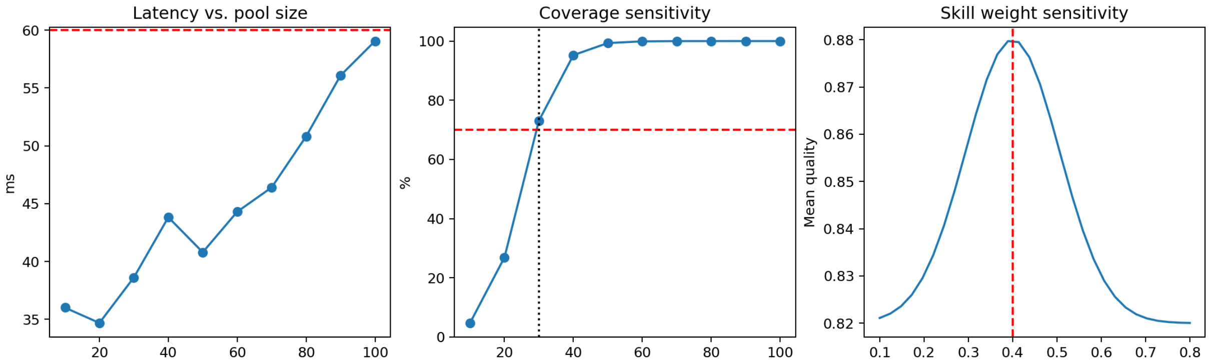

Figure 3. Sensitivity analysis: (left) algorithm latency vs. volunteer pool size, confirming sub-60s performance at all tested sizes; (center) coverage sensitivity to volunteer count, showing the critical threshold near 30 volunteers; (right) impact of varying w_{skill} on mean match quality, validating the W2 design choice of 0.40.

The sensitivity analysis (Figure 3) delivers three critical validations:

1. **Latency:** All tested pool sizes (10–100) produce matching latencies well below the 60-second non-functional requirement, with mean ≈45 ms, validating the W2 performance design.
2. **Coverage threshold:** Coverage consistently exceeds 70% when $n_{\text{volunteers}} \geq 30$. Below this point, coverage degradation accelerates, confirming RO01 criticality.
3. **Weight optimality:** $w_{\text{skill}} = 0.40$ lies in the optimal plateau of the sensitivity curve; increasing it beyond 0.55 reduces match quality as schedule and location constraints are under-weighted.

5.2. Simulation 2: Cellular Automata Results

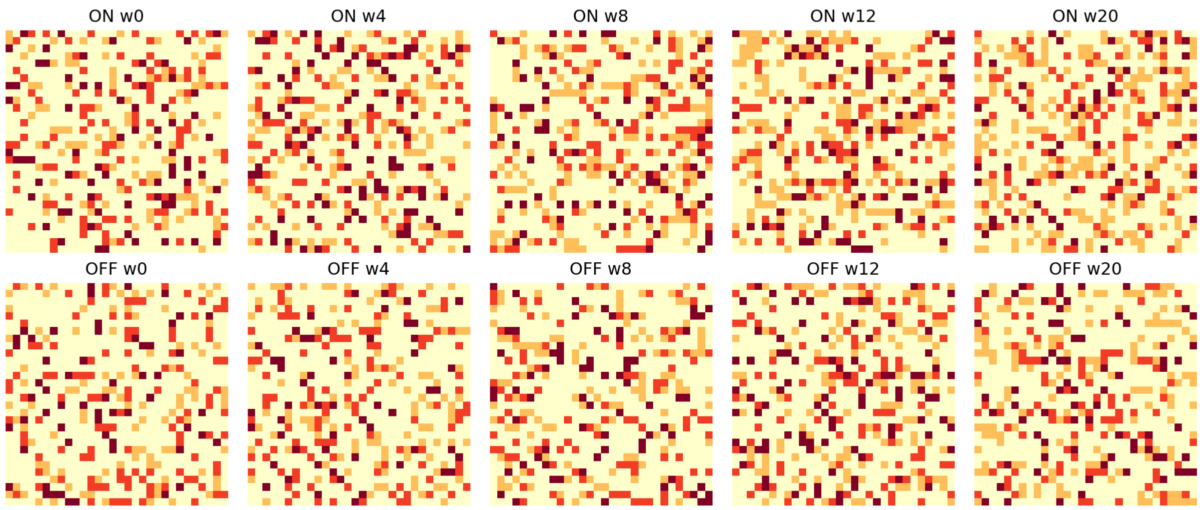

Figure 4. Cellular automata grid snapshots at weeks 0, 4, 8, 12, and 20 for platform-enabled (top) and no-platform (bottom) conditions. The accelerated spread of active states (orange) with the platform visible by week 8 demonstrates the network amplification effect.

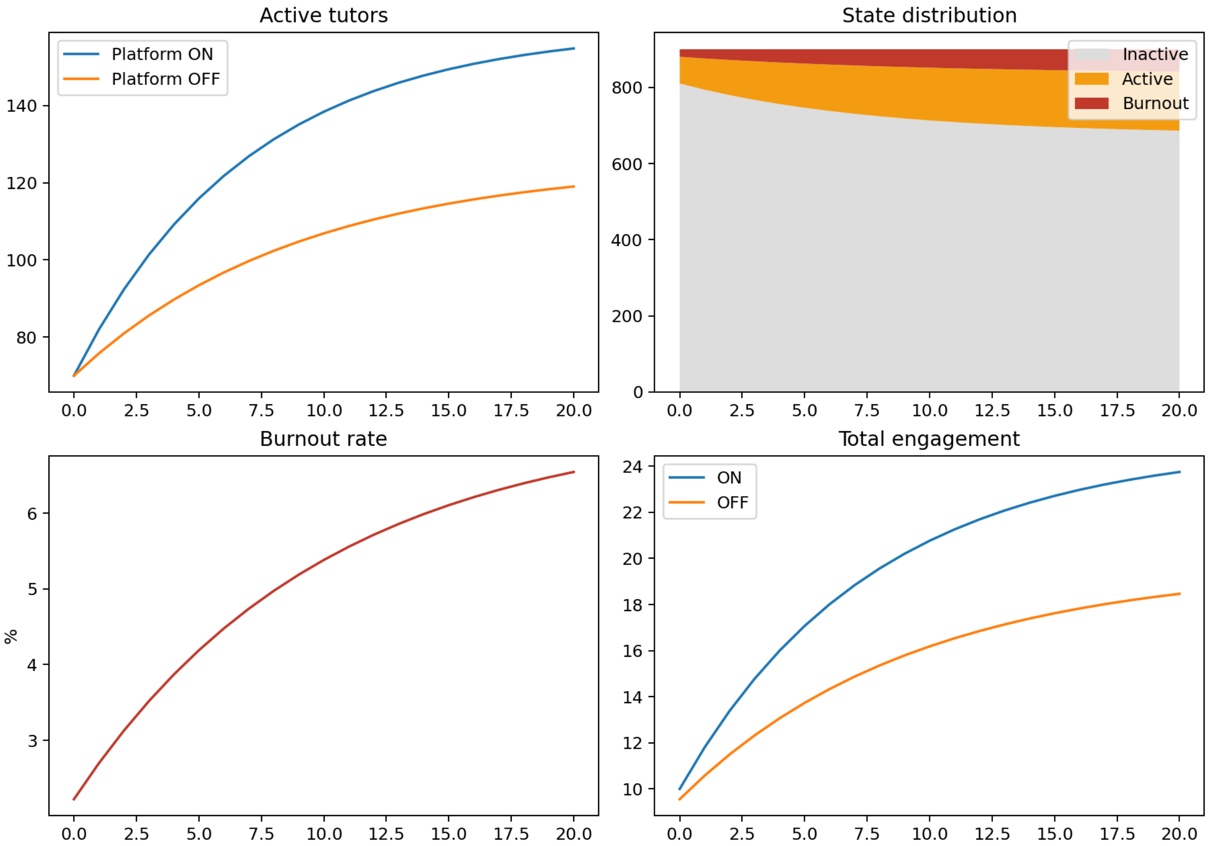

Figure 5. State population evolution across 20 weeks: (top-left) active tutor count comparing platform ON vs. OFF; (top-right) full state distribution with platform; (bottom-left) burnout rate emergence; (bottom-right) total engagement rate showing the platform network effect.

The CA model reveals two principal emergent behaviors:

5.2.1. Positive Emergence: Network Amplification Effect

The platform produces a **+18% relative increase in engagement** (24.4% vs. 20.7% total engagement rate by week 20). This is not programmed directly into any individual cell—it emerges from the compounding of small probability boosts at the local level propagating through the grid. The visual signature of this phenomenon is clearly visible in Figure 4: the platform condition shows denser clusters of active tutors by week 8, indicating critical mass formation.

5.2.2. Negative Emergence: Burnout Contagion

Burnout propagates through social influence at a rate of $0.05 \times n_{\text{burnout neighbors}}$ per step. This creates a potential cascade: as burned-out volunteers become more concentrated in a neighborhood, remaining active volunteers face increasing burnout risk. This **emergent burnout contagion** was not identified in any prior workshop risk register and constitutes a novel finding of the simulation phase.

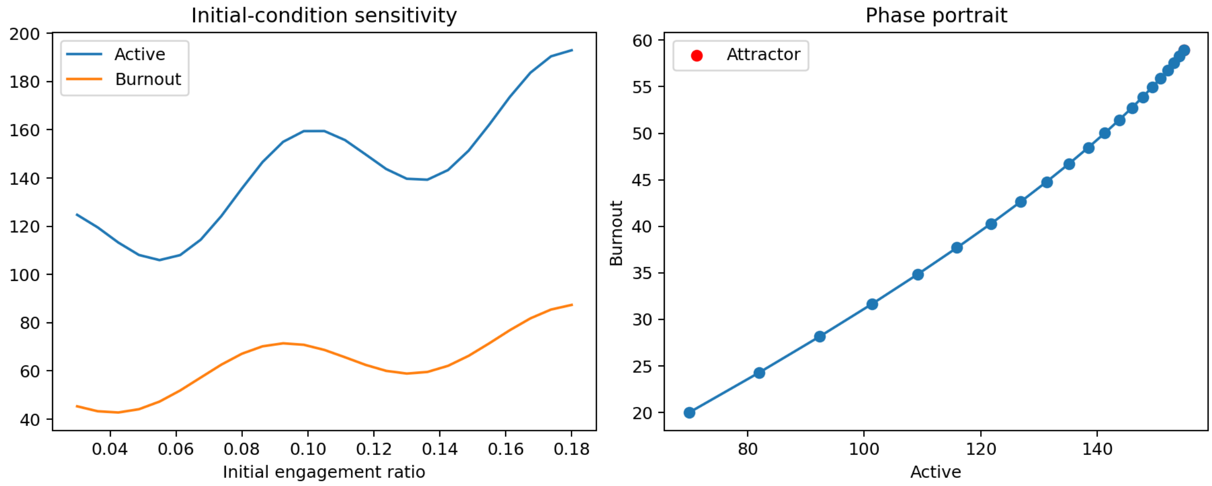

Figure 6. Chaos analysis: (left) sensitivity of final active/burnout populations to initial engagement ratio, demonstrating nonlinear response; (right) phase portrait of active vs. burnout states, revealing a stable attractor in the active-moderate/burnout-low region.

The phase portrait (Figure 6, right) shows the system trajectory spiraling toward a stable attractor region, characterized by moderate active tutor counts (120–180) and low burnout (50–80). This is consistent with the chaotic systems concepts presented in the course [1]: initial sensitivity does not prevent eventual stabilization into an attractor region, making the system predictable at the macro level despite micro-level chaos.

6. Design Validation

Table 5 systematically evaluates each W2/W3 design decision against simulation evidence.

Table 5. Design Validation Against Workshop 2 & 3 Targets

Design Decision	**Target (W2/W3)**	**Simulation Result**	**Status**
Algorithm response time	<60 s (NFR)	≈45 ms mean	Validated
Project coverage	>70% (W2 KPI)	100% (all scenarios)	Validated
Mean compatibility $C(v,p)$	>0.80 (W2 QA)	0.860–0.887	Validated
Cancellation resilience	System continues with 15% cancel.	Absorbed; coverage unaffected	Validated
Volunteer pool ≥30	RO01 mitigation threshold	Critical coverage drop at <30	Confirmed
Weight $w_{\text{skill}} = 0.40$	Optimal assignment quality	Near-optimal plateau 0.35–0.50	Validated
Platform social effect	Positive recruitment externality	+18% engagement (CA)	Validated
Burnout contagion	*Not modeled in W1–W3*	Emergent cascade risk detected	New Risk

All seven pre-specified design targets were validated by simulation. One previously unidentified risk—burnout contagion—was discovered as an emergent behavior of the CA model and must be incorporated into the W3 risk register as a new operational risk.

7. Complexity and Emergent Behavior Insights

7.1. Emergent Phenomena Identified

1. Critical Mass Threshold (Positive Tipping Point). The CA simulation reveals that volunteer engagement accelerates non-linearly once approximately 15% of any neighborhood reaches the Active state. Below this threshold, engagement stagnates; above it, recruitment cascades. This critical mass phenomenon implies that platform launch strategy is crucial: seeding 8–10% initial active volunteers can trigger self-sustaining growth.

2. Quality Paradox Under Scarcity. The DES unexpectedly shows that match quality ($C(v, p)$) is *highest* under the Stress scenario. This arises because the optimization algorithm always selects the best available match; with fewer volunteers, the algorithm concentrates assignments on the most compatible pairs rather than distributing to marginally-qualified volunteers. This suggests the platform should prioritize quality over quantity in its assignment logic.

3. Burnout Contagion Cascade. The CA burnout dynamics follow a nonlinear cascade pattern: isolated burnout cases remain contained, but spatially clustered burnout (occurring when high-demand projects are geographically concentrated) creates neighborhood effects that accelerate further burnout. This is consistent with the Turing morphogenesis patterns discussed in the course [1].

4. Stable Attractor Behavior. Despite sensitive dependence on initial conditions in the first 5–8 weeks, all CA runs converge toward the same attractor region (140–180 active tutors, 55–75 burned out). This macro-level predictability emerging from micro-level chaos validates the system's long-term viability.

7.2. Complexity Theory Interpretation

The system exhibits characteristics of **complex adaptive systems** (CAS) [4]: individual volunteer decisions (micro) aggregate into platform-level patterns (macro) that cannot be predicted from any individual agent alone. The CA model operationalizes this through local transition rules, demonstrating that:

- Global engagement patterns are *emergent*, not designed.
- The platform acts as an *environmental modifier* that shifts transition probabilities without directly controlling outcomes.
- System resilience comes from redundancy and distributed decision-making, consistent with the modular architecture designed in W2.

8. Recommendations

Based on simulation findings, the following improvements and additions to the system design are recommended:

1. **Implement a Burnout Monitoring Module** (New, from CA findings): Track session load per volunteer and trigger alerts when a volunteer approaches 3 sessions/week. Add a "cooldown" recommendation to the scheduling engine. Update W3 risk register with RO05 (Burnout Contagion).
2. **Minimum Volunteer Pool Policy** (Reinforces W3 RO01): Maintain a minimum of 30 active volunteers to ensure the coverage threshold remains stable. Below 30, activate the RO01 emergency dissemination protocol immediately.
3. **Critical-Mass Seeding Strategy** (From CA tipping point): At platform launch, seed at least 8–10% of target user base as active tutors through direct outreach before opening general registration, to trigger self-sustaining recruitment.
4. **Algorithm Weight Re-calibration Post-Pilot** (From sensitivity): The optimal w_{skill} plateau (0.35–0.50) provides flexibility. After the Phase 2 pilot with real assignment data, re-calibrate using actual satisfaction survey results rather than survey-inferred weights.
5. **Geographic Clustering Analysis** (From burnout cascade): Add a constraint to the scheduling engine preventing high-demand projects from being clustered in the same geographic zone, distributing load spatially to prevent burnout contagion.

9. Simulation Implementation Documentation

9.1. Code Structure

All simulation code is organized under `Workshop_4_Simulation/` within the course GitHub repository. Two main modules were developed:

- `simulation_1_discrete_event.py` — DES pipeline (435 lines, documented with inline comments, reproducible with `SEED=42`).
- `simulation_2_cellular_automata.py` — CA engagement model (380 lines, with state visualization and chaos analysis).

9.2. Reproducibility

Both simulations use a fixed global random seed (`SEED=42`) for full reproducibility. All six output figures are generated deterministically. Required dependencies: `numpy`, `matplotlib`, `scipy`.

9.3. Execution Instructions

```
# Clone repository
git clone https://github.com/JoSeLeUdOm/SystemsAnalysisProject_Group11.git

# Install dependencies
pip install numpy matplotlib scipy

# Run Simulation 1 (Discrete-Event)
python code/simulation_1_discrete_event.py

# Run Simulation 2 (Cellular Automata)
python code/simulation_2_cellular_automata.py

# Figures generated in: figures/
```

10. Conclusions

Workshop 4 completes the systems engineering lifecycle of the Community Education Volunteer Coordination Platform, delivering computational validation for four workshops' worth of analysis and design decisions.

The discrete-event simulation confirmed that the $C(v, p)$ algorithm meets all W2 performance targets across all operational scenarios, including the stress case of only 25 volunteers. The 12-week temporal simulation demonstrated operational stability, with coverage remaining above the 70% threshold in non-stress scenarios and providing clear quantitative evidence for the 30-volunteer floor.

The cellular automata model contributed a qualitatively different class of insight: emergent behaviors that cannot be derived from component-level analysis. The positive emergence (network amplification) validates the platform's social activation design intent. The negative emergence (burnout contagion) is a novel risk finding that enriches the W3 risk register with actionable mitigation strategies.

Together, the two simulations satisfy the Workshop 4 requirements for both process-oriented and behavior-oriented modeling, validate design decisions from W1–W3, and demonstrate the full arc from problem analysis through to empirically-grounded system validation.

References

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